



**AIAA 92-3989**

**Hypersonic Flight Testing**

W. E. Williamson

Sandia National Laboratories

Albuquerque, NM

**AIAA 17th**  
**Aerospace Ground Testing Conference**  
July 6-8, 1992 / Nashville, TN

## HYPERSONIC FLIGHT TESTING

W. E. Williamson\*  
 Sandia National Laboratories  
 Albuquerque, New Mexico

**Abstract**

As part of the design process for a hypersonic vehicle, it is necessary to predict the aerodynamic and aerothermodynamic environment for flight conditions. This involves combining results obtained from ground testing with analytical modeling to predict the aerodynamics and heating for all conditions of interest. The question which always arises is, how well will these models predict what is actually seen in a flight environment? This paper will briefly address ground-testing and analytical modeling and discuss where each is appropriate, and the associated problems with each area. It will then describe flight test options as well as instrumentation currently available and show how flight tests can be used to validate or improve models. Finally, several results will be shown to indicate areas where ground testing and modeling alone are inadequate to accurately predict hypersonic aerodynamics and aerothermodynamics.

**Introduction**

Prior to the development of hypersonic flight vehicles, a significant investment is made in ground testing (wind tunnel, arc jet, etc.) and analytical modeling (CFD). Since the vehicle design in terms of thermal protection, control, cooling, etc., is dependent upon these predictions, it is necessary that we have

\*Dept. Mngr., Hypersonic Devel. Dept.  
 Associate Fellow, AIAA

This work performed at Sandia National Laboratories supported by the U.S. Department of Energy under contract DE-AC04-76DP00789.

Copyright ©1992 by the American Institute of Aeronautics and Astronautics, Inc. All right reserved.

some confidence in these results in order to produce an optimal design. Based on previous flight test experience, we know that surprises usually occur during the first few flight tests of any vehicle. One option to reduce risks and help eliminate surprises is to develop small unmanned vehicles which can be flown in the hypersonic environment to validate models. This paper will describe some of the options available in the small-scale flight testing area and the relationship between the results obtained by flight testing, ground testing, and simulation. In particular, it hopes to show that ground testing and modeling alone are inadequate to predict aerodynamics and aerothermodynamics with a high degree of accuracy in the hypersonic environment.

The order of the presentation will be to first discuss ground testing and analytical modeling and some of the issues relating to this data. Next flight test options and instruments available to make measurements during the flights are discussed. These measurements are then related to preflight predictions and comments about the relationship between the preflight predictions and actual measurements will be made.

**Vehicle Design Issues**

The first step in the design process for hypersonic vehicles is the generation of a set of performance specifications. A nominal design is created which includes the external geometry. Ground testing and analytical models are used to predict aerodynamics and aerothermodynamics. These results are then used to predict vehicle performance and subsequently iterate toward the desired configuration. In some instances, some prototype flight tests are permitted in order to verify prediction techniques. The flight test results are used to improve the prediction models and hence move toward the optimal

configurations with higher confidence than if only ground testing is available.

Approaches for studying aerodynamic and aerothermodynamic results on a hypersonic vehicle are well known. They include wind tunnel and arc jet ground test numerical simulation techniques (including CFD and approximate solution techniques), and vehicle flight tests. All of these are used to predict aerodynamic and aerothermodynamic results for a hypersonic vehicle and hence iterate toward a final vehicle design.

With a large number of ground test facilities available and with the growth of CFD during the last few years, why do we need flight tests? It would seem that we could predict quite accurately aerodynamic and aerothermodynamic effects on almost any shape at any flight condition.

In addition, flight testing is very expensive, takes money away from the development of the actual vehicle, and requires long preparation times to design, fabricate, and flight-test qualify all of the required hardware and software. Many things can go wrong in flight testing. Occasionally, a booster failure carrying an expensive payload may result in the cancellation of a program, even though the failure is unrelated to the experiment. There seem to be many more bad things than good which can occur in flight testing.

On the other hand, flight testing is the only true measure of predicted aerodynamics and aerothermodynamics. If accurate predictions are required for vehicle design, only flight testing can identify the appropriate conditions and produce results required for near optimal vehicle design. Results obtained from flight tests which show this is correct, are shown later.

On the other hand, there are also issues relating to ground testing and modeling. Wind tunnels, for example, always use subscale models at limited flow conditions. The result of this is that the data acquired in a wind tunnel is almost never at the desired flight Reynold's Number. In addition, it is often not possible to test at the correct Mach number, or certainly at all Mach numbers of

interest. Most high Mach facilities are not continuous flow tunnels and data is limited to very short test times. This limits control effectiveness studies since only data for a limited number of control deflections is affordable. It is usually too expensive and time-consuming to acquire all the desired data at high Mach ( $M > 12$ ). The standard tunnel problems relating to sting effects and tunnel flow properties also exist.

Fortunately, there are some positive attributes of tunnel testing. In continuous flow tunnels, it is extremely efficient to study angle of attack, roll angle, and control surface deflection effects. A great deal of data can be acquired for these effects, even though it is usually Mach and Reynold's Number limited. It can also be used to study general geometry and configuration changes and show resulting hypersonic trends.

The test conditions for a vehicle designed to fly in the hypersonic region are shown in Figure 1. This is a very large range of tests designed to predict aerodynamic and aerothermodynamic results on a hypersonic vehicle and is representative of most tunnel test conditions in the U.S. It is plotted versus altitude since it is sometimes more descriptive than Reynold's Number. For the higher Mach Number data, the situation is not as good as it appears. These are blow-down tunnels often utilizing helium. Costs prohibit the collection of a large amount of data and most of it must be corrected to conditions appropriate for air. In all instances, real gas effects are ignored in the data and it is obvious that some desired flight conditions cannot be achieved. The vehicle considered in Figure 1 is 8 feet long. For a vehicle 200 feet long, the corresponding altitudes are much higher and it is not possible to acquire data at desired flight altitudes near 100,000 feet.

By not being able to test all conditions, we risk missing a flow condition that could create a complex wing-body shock structure and heating may be amplified by several orders of magnitude in this region. Real gas effects are not present in the tunnel and may affect results. The inability to test at all flight conditions may cause critical areas of the

flight to be ignored.

There are also issues relating to the computational fluid dynamics (CFD) area. The CFD computer programs which have been developed range from inviscid flow field codes and associated boundary layer codes, to parabolized Navier Stokes or full Navier Stokes codes. These are standard codes for simple ballistic reentry vehicle shapes but require a significant effort to generate an adequate grid structure for complex shapes. In addition, most of these codes have a few parameters which the user must select prior to computing answers. In general, these parameters are selected by picking values which cause the CFD results to match wind tunnel results. If no wind tunnel data is available, or the user does not have a great deal of experience with the code, the CFD results often are very poor. In addition, the CFD tools often do not work for the complex geometry associated with maneuvering shapes. If the user is entirely dependent on CFD, they may be left in the position of not being able to compute results for all of the vehicle shapes and vehicle attitudes of interest. The simpler codes, in the HABS class, almost always produce an answer but do not model the flow field correctly. The choice is often between a computed but questionable answer and a complex flow field code which may not compute the answer of interest.

Fortunately, however, CFD does supply a tool to predict Mach and Reynold's Number trends where no wind tunnel data is available. By adjusting parameters to match ground test data on either side of the predicted value, the user is generally confident of results obtained. It provides an extremely useful tool for studying the effects of Reynold's Number, Mach Number, real gas and transition on results.

### Flight Test Options

The comments in the previous section have been intended to point out the limitations of ground testing and CFD for designing a vehicle to fly in the hypersonic regime. This next section will describe several flight test options that are available. These options

consist of a variety of flight test ranges as well as booster options which can be used to place a small vehicle (9 - 25 ft) in the appropriate flight region to study hypersonics. Several of the currently available booster options include the Shuttle, MX, Minuteman, Polaris, Pegasus, and small sounding rockets. Scaled models of hypersonic vehicles can be flown from any of the booster system options and provide a wide variety of flight conditions including large Mach and altitude variations.

A variety of small rockets are shown in Figure 2. These are systems flown at Sandia, but are indicative of the sizes flown by other companies. These vehicles can launch payloads from tens of pounds to hundreds of pounds to hypersonic flight conditions. They can provide relatively cheap data on an actual flight model prior to full scale vehicle development.

Figure 3 shows the STRYPI booster system designed at Sandia and compares it with the Polaris (STARS) booster. Sandia is currently developing the capability of launching the STARS missiles from the Kauai Test Facility. This will provide a new flight test option for hypersonic flight testing. It provides the capability of boosting a payload in the 500 lb. to 900 lb. category to between 18,500 ft/sec and 17,000 ft/sec with -55 deg to -35 deg flight path inclinations.

There is a large variety of flight test location options. Shown in Figure 4 are possible launch sites in the Pacific Ocean. Significant ground radar and telemetry capability exists on both Kwajalein and Kauai. The Air Force regularly flies ballistic reentry vehicles from Vandenberg into the Kwajalein Atoll. The selection of launch and reentry areas depends on radar and telemetry coverage required, range safety issues, and required ground support.

One typical option is shown in Figure 5. This payload was launched from Wake Island and reentered near Kwajalein to take advantage of the radar and telemetry capabilities at Kwajalein. The STARS booster provides the possibility of flying from Kauai to Kwajalein in a similar manner.

### Preflight Predictions

Prior to a flight test, a significant amount of work is done preflight in predicting performance of the vehicle. This includes not only the aerodynamic and aerothermodynamic environments, but also the atmospheric, mass, structural, and material properties. Once these are predicted, trajectories, indepth temperatures, and structural response may be predicted. This list of things is shown in Figure 6. All of this information is required to predict vehicle performance.

### Offboard Instrumentation

Due to the large investment in flight testing, it is then desirable to make as many measurements as possible during the flight to help verify predictions or explain any modeling inaccuracies. The instrumentation should measure as directly as possible the things that have been predicted. This is often not possible and it is often necessary to infer predictions from related but not direct measurements. A well-instrumented range, such as the one shown in Figure 7, is a requirement. This range includes radar optical sensors and telemetry receiving units. In remote areas, mobile facilities or aircraft may be used to supply these functions.

Measurements fall into three groups. These include atmospheric properties measurements, offboard vehicle related sensor measurements, and onboard vehicle related measurements telemetried to the ground or stored on tape and retrieved post-flight. Atmospheric measurements determine wind conditions, atmospheric density, temperature, and pressure profiles as a function of altitude. They are extremely important, since at some latitudes, atmospheric properties vary significantly. Uncertainty in these quantities can result in poor comparisons of predicted and measured results. In general, balloons and sounding rockets are launched to determine atmospheric properties and winds both near the booster launch area and near the reentry area.

Figure 8 shows the Sandia-operated Kauai Test Facility. It shows launchers, assembly

buildings, radar and telemetry antennas, and radar to track balloon launches. Additional facilities are described below.

The two X-band antennas shown in Figure 9 are used to track balloons just prior to launch. Ballistic rockets then have their launcher settings adjusted to compensate for winds so that they will still achieve proper reentry conditions.

Shown in Figure 10 is a typical optical tracking telescope used to observe launches and occasionally reentry. High quality photo coverage is used to help identify causes of failures during flight as well as provide postflight trajectory information.

Figure 11 shows a mobile X-band tracking radar and S-band telemetry receiver. They could be located as close as range safety permits to the reentry area to acquire "close in" data.

### Onboard Instrumentation

Once predictions on performance are made, it is desirable to select onboard instruments which measure as directly as possible the predictions. Figure 12 summarizes the types of instrumentation which could be flown on a reentry vehicle. It includes a large number of instruments to measure dynamic, aerothermodynamic, aerodynamic, structural, and material performance. Note that none of this information measures flow field properties or surface heating rates directly. It measures the effect through dynamic instruments or temperature measurements. Flow field results must be inferred through dynamic modeling or in-depth temperature modeling.

The next few figures show pictures of several of the instrument types flown on-board. They provide an indication of the sizes of various instruments which could be flown on small unmanned reentry vehicles.

Figure 13 shows a nosetip balance. The nosetip is mounted to one end and the remainder of the vehicle to the other end. Loads on the nose can then be measured directly during flight. Similar strain gauge

combinations can be used to measure loads on control surfaces during flight. These gauges measure the forces and moments on a nose or surface. Again, this is not a direct measurement of the flow field but the effect of the flow on the forces and moments on a potentially ablating surface.

The Sandia Aerospace Computer (SANDAC) and Roll Stabilized Inertial Measuring Systems (RIMS) which can be used to navigate, determine attitude, and generate autopilot commands are shown in Figure 14. The SANDAC is silver on top, RIMS is black and centered on the second level. To provide a perspective on the size, SANDAC is about 8" long. The entire collection weighs about 25 lbs. This is indicative of the size and weight requirements for a small vehicle computer and platform which can be used to provide a large amount of onboard information, including angle of attack information for post-flight analysis.

A typical Sandia telemetry system (about 15" long) is shown in Figure 15. It consists of accelerometers, gyros, signal conditioners, and commutators. Information is collected here and radiated through windows on the side of the vehicle to ground stations.

A typical thermocouple with several mounting heatshield material plugs are shown in Figure 16. The plugs are 1" long. The wires lie along the side of the plug, and the actual sensor extends at right angles into the plug. A typical plug will contain 3 thermocouples at different depths, with all the sensors aligned over the center of the plug. Care is taken to align the tape wrap of the plug with that of the heatshield.

The photodiode is a relatively new instrument flown on reentry vehicles. It consists of a small (0.5 in diameter) glass window and a small sensing device behind it. It can be mounted through the heatshield as shown in Figure 17.

The photodiode works by detecting light intensity exterior to the window. This is described in Figure 18. When the flow goes from laminar to turbulent, there is a significant change in intensity. The

photodiode observes this and thus provides an excellent transition indicator. By having several over the surface, an approximate determination of the transition boundary can be made.

Several instruments can be used to measure the amount of ablation in a nosetip or heatshield material. Examples of these are shown in Figure 19.

The radioactive gauge measured the effective radiation from a strip implant. The radioactivity, which decreased as the material ablated away, was observed. Implanting the radioactive material in the nosetip required drilling holes in the nosetip. This created the uncertainty of the effect of the holes on the nosetip ablation.

The acoustic gauge does not require the radioactive material or implanting procedure, and hence seems to be a better choice. These gauges cannot provide detailed ablated geometry but do provide an average recession length measurement.

An important area of present research is nonintrusive instrumentation. There is a lot of current research aimed at measuring flow fields directly with instruments which do not extend into the flow. This work could become very important in the future since it would provide a direct flow-field measurement to compare with CFD.

### Postflight Analysis

After a flight has occurred and all the data have been collected, it is desirable to compare the predictions to the data and improve prediction techniques. The first step in the comparison is to produce the best estimate of the actual trajectory. This will involve combining atmospheric measurements of temperature, density, and pressure with radar tracks, optical tracks, and onboard dynamics-related instruments. There are a large number of schemes for doing this, ranging from fairly crude (filter range, azimuth, and elevation data from a radar and differentiate this data for velocity) to very elaborate (integrate accelerometer and gyro data in a 6-DOF code; adjust scale factors

and initial conditions using a Kalman filter to best match radar and optical data). The end result is a tabulation of position and velocity versus time. Since atmospheric properties are also known, the dynamic pressure, Mach Number, and Reynold's Number can also be computed.

Once a trajectory is known, lateral dynamic instruments can be used to estimate lateral aerodynamic coefficients such as  $C_m$  or  $C_N$ . Note that this does not validate flow-field models, but is a check on the integrated effect of the flow-field. Predictions of these coefficients can agree with flight results and the flow-field code can still be incorrect. Techniques for estimating these coefficients are usually in the class of weighted least squares fits, Maximum Likelihood Estimators, or Kalman filters. For a spinning vehicle, a standard estimation procedure uses a 4-state model consisting of pitching rate, yawing rate, body fixed pitch velocity, and body fixed yaw velocity. These four ordinary differential equations can be numerically integrated if the spin rate, dynamic pressure, and total velocity are known from the trajectory estimation. If the body is not spinning, pitch and yaw are usually separated and a 2-state model is used for aerodynamic coefficient estimation.

The typical procedure for estimating aerodynamic coefficients involves assuming they are constant over short time intervals, fitting a short segment of data to obtain the estimates, and marching the interval forward in time until all data is processed. Figure 20 shows a typical fit of lateral accelerometer data for a ballistic reentry vehicle and is typical of the accuracy obtained in the fit. Figure 21 shows the associated fit of the gyro data.

If more complex modeling is desired, polynomial or elaborate models with parameters can be used to fit the data. In either event, the result is a postflight estimation of the aerodynamic coefficients during flight.

As an example of this procedure, Figure 22 shows the predicted and estimated values of  $C_m$  for a small ablating reentry vehicle. It can

be seen that there is not good agreement between the predicted and post-flight estimated value. In this case, the high altitude values were predicted with an inviscid flow field code. In flight, the Reynold's Number is low at high altitude and the inviscid assumption is not correct. Low altitude differences are attributed to nosetip shape change. These differences cannot be accurately predicted, and require several flight tests of the same material before the behavior can be accurately modeled. Figure 23 is the corresponding plot for  $C_N$  again showing significant differences between prediction and flight derived estimates. The corresponding static margin plot is shown in Figure 24. In this case, the nosetip ablation increased static margin late in the flight. In other instances, it has been observed to decrease static margin. These effects were not predicted preflight. The heatshield blowing and nosetip ablations have never been predicted well for new materials prior to flight. These are some of the things which require flight testing to validate performance.

After the trajectory, attitude information, and atmospheric properties are known, heating comparisons can be made between predictions and inflight measurements such as thermocouples. The comparison is again indirect, since heatshield materials properties must be known and predictions of indepth temperatures are compared with flight measures. When ablation occurs, it is difficult to determine the external geometry, and this uncertainty contributes to inaccuracies in the heating comparisons.

Figure 25 shows a typical plot of thermocouple data and predictions. This vehicle used a standard carbon phenolic heatshield and with all of the knowledge about the material, the agreement is very good. This is usually not true when a new heatshield material is used.

Another critical element for the heating prediction is the transition criteria. This is extremely difficult to predict on a new configuration until flight test data is acquired. Its effect is extremely important, however, in a thermal protection system design. Whether the flow remains laminar or becomes

turbulent will result in a significantly different heatshield design.

Figure 26 shows photodiode and thermocouple data on a maneuvering vehicle. The photodiode was used to verify that the vehicle was on the verge of turbulent flow and in fact jumped back and forth between laminar and turbulent flow. The actual flight conditions are shown in Figure 27 and are compared with two standard techniques for predicting transition. It can be seen that the flight data is not in agreement with either prediction technique.

Figures 28 and 29 show the same two standard methods of predicting transition shown in Figure 27, the G.E. Low Mass Addition technique and a typical approach used for NASP where  $Re_\theta = 150 M_\infty$  is used. Also plotted are numerous results obtained for ballistic reentry vehicles. The predictions are not good. While these vehicles are ablating and have blowing heatshields, the results are clearly indicative of our inability to predict transition.

The final step in the postflight analysis is an assessment of material ablation and shape change. These results can be seen in recession gauges, nosetip balances, thermocouple burn-through, and in the aerodynamic coefficient estimates. It is often difficult to infer the actual shape, but general comments about the average recession are very important to ensure that margins in heatshield thickness and nosetip length are adequate.

### Summary

In summary, ground testing and prediction techniques are necessary for preliminary design. This data is notorious for having missed some element of the actual flight performance, such as transition. The only way to determine if all elements are considered is to conduct a flight test. It is essential for verifying prediction techniques and providing high confidence in the vehicle design.

There are many options for flight testing including flying small unmanned models on

present booster systems. Many options exist for launch and reentry sites. There is a large source of onboard instruments which can accurately measure things during flight. Finally, there are well-developed procedures for using the measurements to verify models or demonstrate where models are incorrect.





- Figure 6. Preflight Predictions/Measurements**



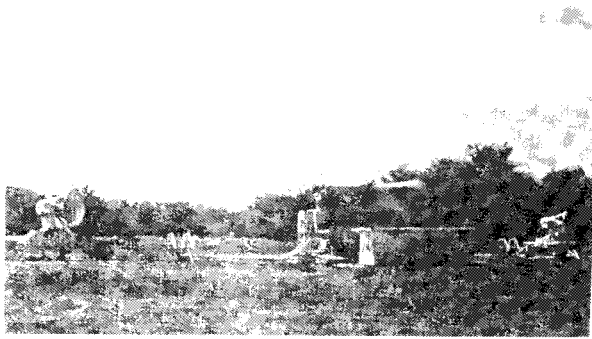


Figure 9. Balloon Tracking Radar

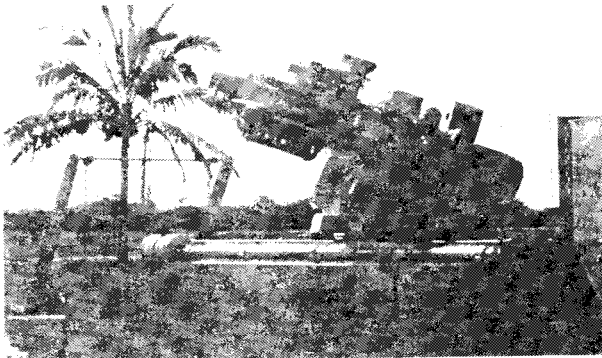


Figure 10. Optical Tracking Telescope

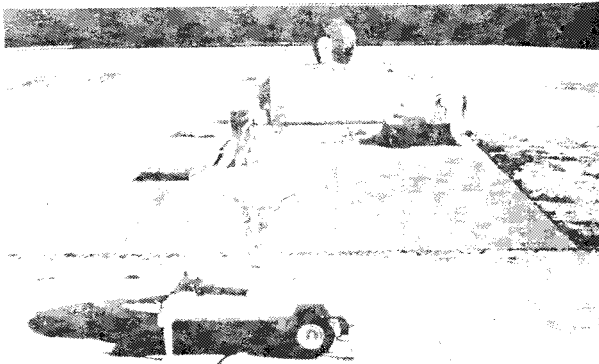


Figure 11. Mobile Radar and Telemetry Receiving

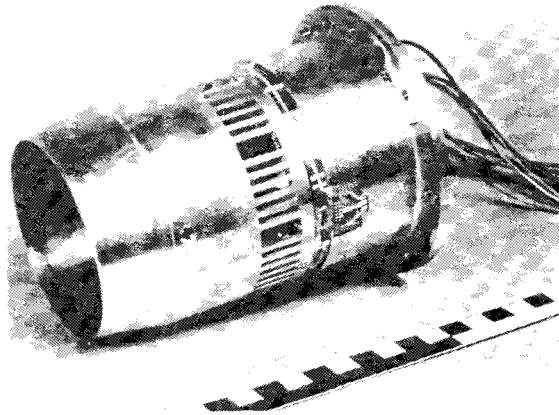


Figure 13. Nosetip Balance

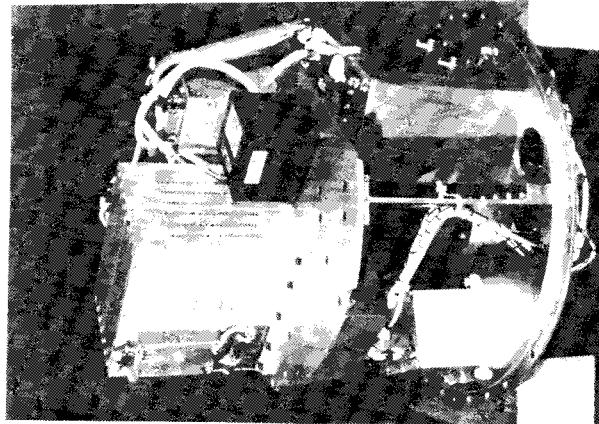


Figure 14. Onboard Computer and IMU

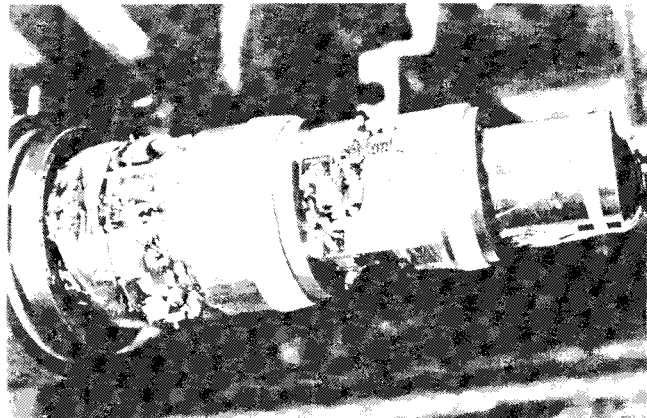


Figure 15. Onboard Telemetry System

|                   | Instrument                   | Measurement Type             |
|-------------------|------------------------------|------------------------------|
| Dynamic           | Accelerometers               | Accel (3 Axis)               |
|                   | Gyro                         | Angle Rates (3 Axis)         |
|                   | Platform-Computer            | Pos, Vel, Angle, Angle Rates |
|                   | Magnetometer                 | Angle                        |
|                   | High Frequency Accelerometer | Vibration                    |
| Aerothermodynamic | Acoustic                     | Transition                   |
|                   | Nosetip Balance              | Force, Moment                |
|                   | Hinge Moment Strain Gage     | Force, Moment                |
|                   | Actuator Pressure            | Force                        |
|                   | Thermocouples                | Temp                         |
| Material          | Photodiodes                  | Transition                   |
|                   | Pyrometers                   | Temp                         |
|                   | Thinistors                   | Temp                         |
|                   | Calorimeters                 | Heat Rate                    |
|                   | Ultrasonic                   | Recession                    |
|                   | Radioactive Strip            | Recession                    |
|                   | Thermocouple                 | Recession                    |
|                   | R.F. Coupling                | Elect Conduct                |

Figure 12. Typical Onboard Instrumentation

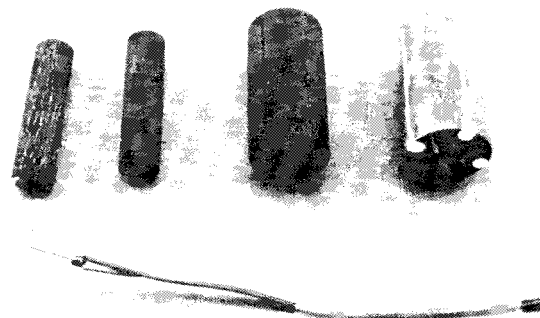
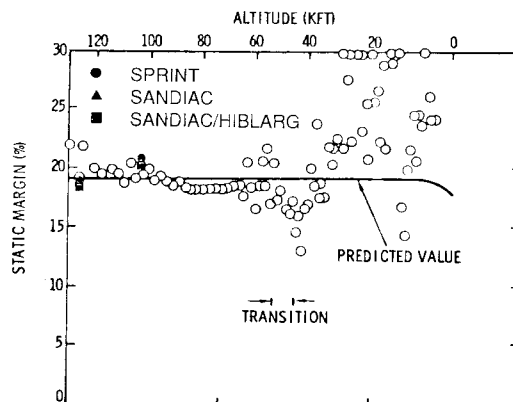
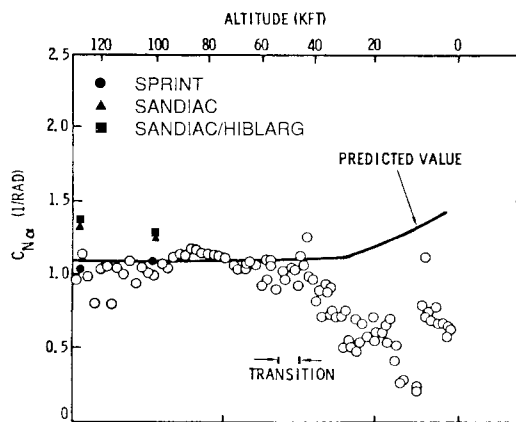
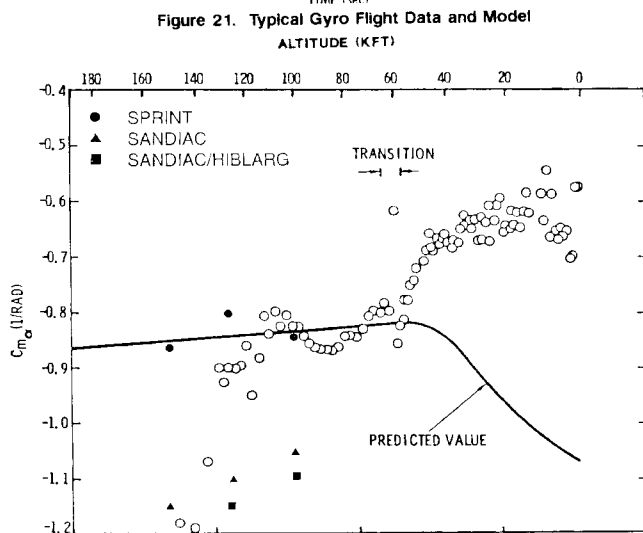
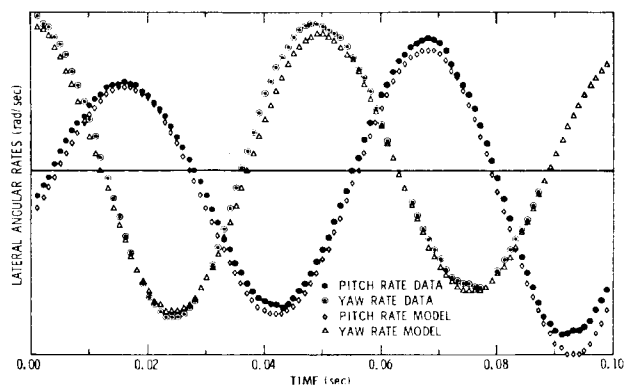
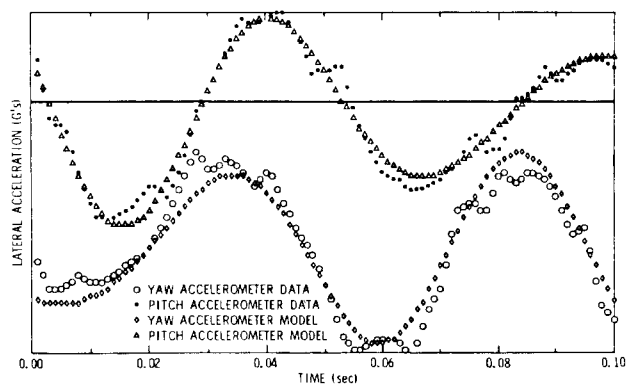
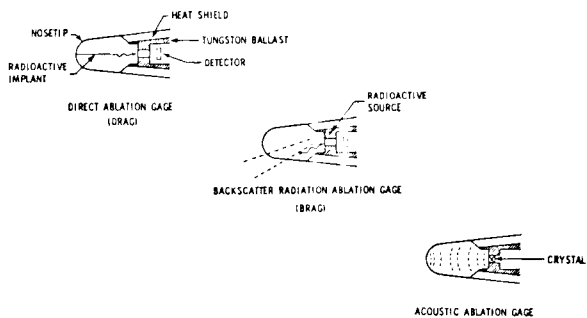
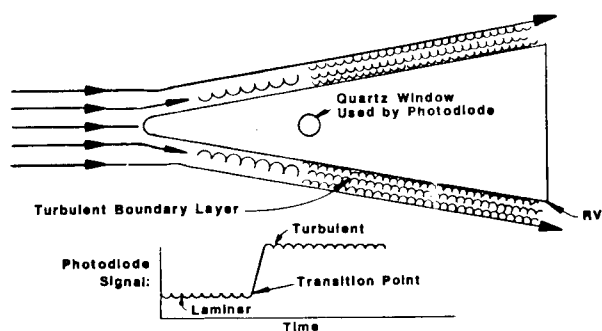
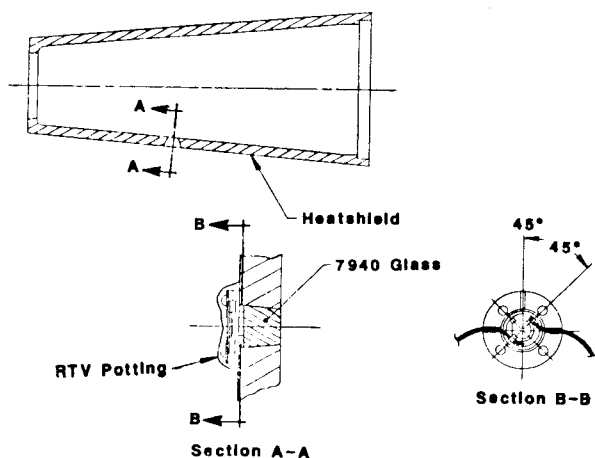


Figure 16. Thermocouple and Mounting Plugs



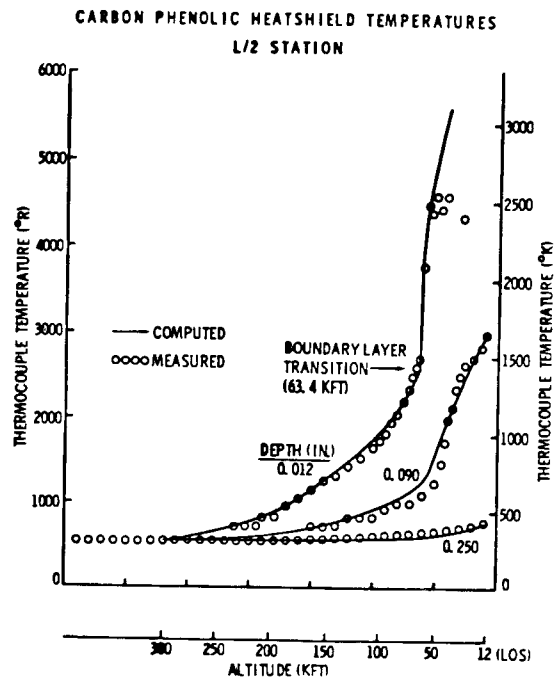


Figure 25. Comparison of Measured and Predicted Temperature

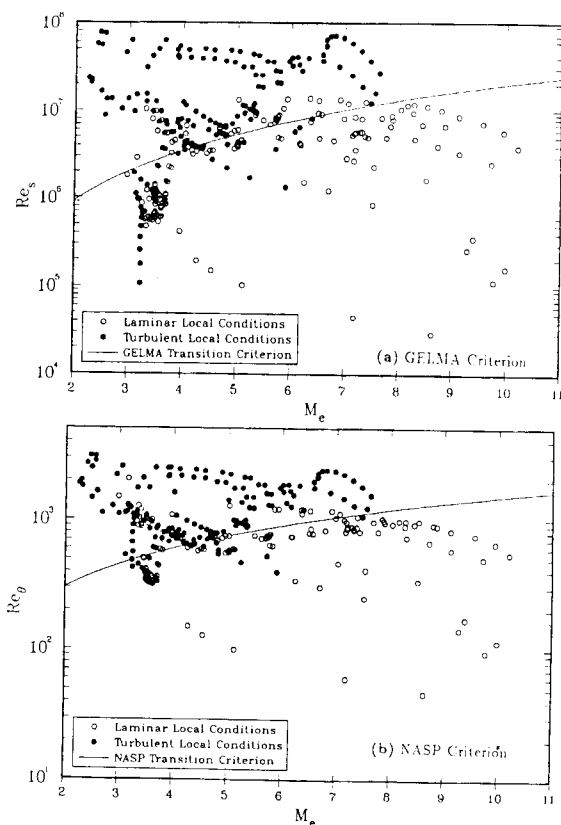


Figure 27. Comparison of Laminar and Turbulent Flow Conditions Compared to NASP Transition Criterion

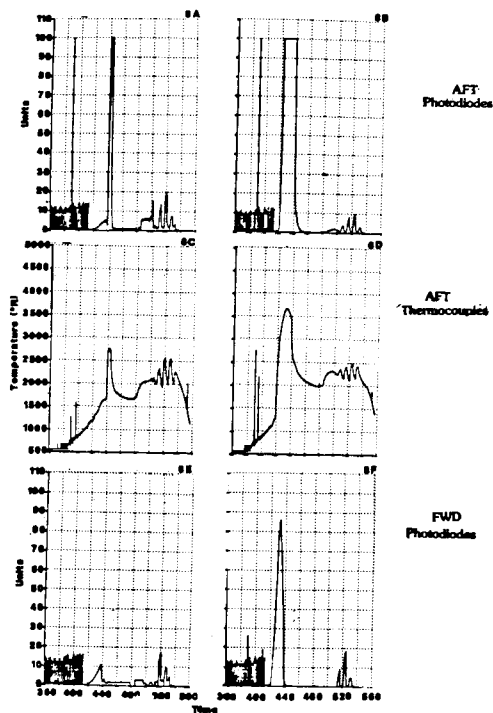


Figure 26. Photodiode and Thermocouple Response Plots

Figure 28. Comparison of GELMA Transition Criterion with Sandia Flight Data

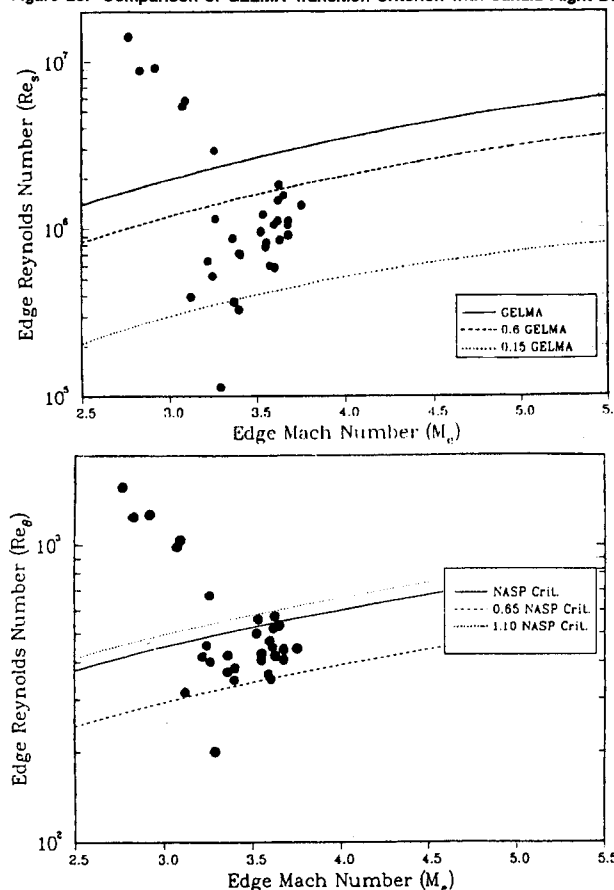


Figure 29. Comparison of NASP Transition Criteria and Sandia Ballistic Flight Data